ASTRONOMY OLYMPIAD 2026

Workshop & Student Preparation Guide

For Junior and Senior students • Unofficial educational resource

Important note about the competition

The supplied poster advertises “NASA Astronomy Olympiad 2026,” while the uploaded syllabus PDFs are titled “International Space Olympiad 2026” and the shared websites include IOAA materials. These are not automatically the same event. Before publishing official claims, confirm the organiser, eligibility, exam format, registration fee, date, and any travel or camp award directly with the event organiser. The practice papers in this pack are original workshop materials and are not official papers.

1. Suggested workshop plan (2 hours)

Time	Session	What to cover
0–10 min	Welcome & orientation	What astronomy is; why the competition is interesting.
10–25 min	Competition briefing	Confirmed organiser, date, eligibility, registration, exam format and rules.
25–45 min	Our place in space	Earth–Moon–Sun, planets, gravity, day/night and seasons.
45–65 min	Stars and galaxies	How stars shine, star colours, Milky Way, galaxy types.
65–75 min	Break / quiz	A short team-based astronomy quiz.
75–95 min	Space science in action	Eclipses, satellites, rockets, telescopes and exoplanets.
95–115 min	Practice questions	Students attempt age-appropriate questions and discuss reasoning.
115–120 min	Next steps	Study plan, resources and Q&A.;

2. Junior study roadmap

Start here: Earth, Moon, Sun, day and night, seasons, the eight planets, gravity and orbits, stars, constellations, comets, asteroids, satellites and rockets.

Then learn: lunar and solar eclipses, Moon phases, tides, the Sun’s layers, sunspots, solar flares, dwarf planets, the habitable zone, star life cycles, galaxies, the Big Bang, exoplanets, Hubble and James Webb telescopes.

How to study: Draw labelled diagrams; explain concepts in your own words; make planet comparison cards; practise 10–15 MCQs at a time; learn the difference between similar ideas (rotation/revolution, star/planet, asteroid/comet).

3. Senior study roadmap

Core concepts: celestial coordinates at an introductory level, Kepler’s laws, orbital periods, gravity, the Sun and fusion, planetary atmospheres, stellar distance and brightness, spectral classes, binary stars, stellar evolution, galaxies, redshift, expansion of the universe, black holes, exoplanet detection and telescopes.

Accessible mathematics: scientific notation, ratios, unit conversion, inverse-square relationships, simple rearrangement of equations, reading graphs and estimating values. Start with conceptual understanding before adding calculations.

Keep advanced topics qualitative: gravitational waves, dark matter, dark energy, neutron stars and cosmology can be introduced through evidence and examples; avoid requiring advanced derivations in this beginner-friendly practice set.

4. Types of questions students should practise

• **Recall:** Which planet is largest? Which galaxy contains the Solar System?
• **Conceptual:** Why do seasons occur? Why does the Moon show phases?
• **Compare:** Star versus planet; comet versus asteroid; optical versus radio telescope.
• **Apply:** Use a simple formula for orbital period or inverse-square brightness.
• **Interpret:** Read a small table or graph of brightness, temperature, or distance.
• **Explain evidence:** How do redshift, exoplanet transits, or star colours tell us about space?

5. Suggested preparation schedule (10 days)

Day	Junior focus	Senior focus
1	Earth, Moon, Sun	Units, scientific notation, gravity
2	Planets and orbits	Kepler’s laws and orbital periods
3	Stars and night sky	Light, temperature and spectra
4	Satellites and rockets	Solar physics and planetary science
5	Eclipses and Moon phases	Distances, parallax and brightness
6	Sun and solar activity	Stellar life cycles
7	Galaxies and universe	Galaxies, redshift and cosmology
8	Exoplanets and telescopes	Exoplanets and observatories
9	Mixed practice questions	Mixed concepts and calculations
10	Mock paper + review	Mock paper + review

6. Recommended resources

• Junior IOAA syllabus: https://ioaastrophysics.org/junior-ioaa/junior-ioaa-syllabus
• IOAA syllabus: https://ioaastrophysics.org/resources/ioaa-syllabus
• Past IOAA problems and solutions: https://ioaastrophysics.org/resources/problems-from-past-ioaa
• International Space Olympiad syllabus page: https://internationalspaceolympiad.com/syllabus/
Use advanced IOAA problems selectively; they are not a difficulty target for this introductory workshop.

7. Instructor advice

Prioritise curiosity, clear explanations and confidence. Ask students to explain why an answer is correct, not just choose a letter. Use diagrams and demonstrations. For any official competition detail, rely on written confirmation from the event organiser rather than assuming details from another astronomy olympiad.

Prepared as an educational workshop resource. The model papers are original practice questions and do not represent the official exam format.